

Public Perceptions of Immigration in Spain: A Quantitative Analysis of Attitudes

Aswin Sarasan

Universitat Autònoma de Barcelona (UAB)

Universidad Pública de Navarra (UPNA)

Universitat de les Illes Balears (UIB)

Master Thesis

MSc. in Management, Organization and Business Economics

Supervisor: **Camilla Civardi**

Universitat Autònoma de Barcelona

Period: 1st of June 2026 until 25th of August 2026

Contents

Abstract.....	3
1. Introduction.....	4
2. Literature Review and Hypothesis Development.....	5
2.1 Understanding Public Attitudes Towards Immigration	5
2.2 Education and Immigration Attitudes	6
2.3 Political Ideology and Immigration Attitudes	7
2.4 Economic Insecurity and Immigration Attitudes.....	8
2.5 Intergroup Contact and Immigration Attitudes.....	9
2.6 Summary of Hypotheses.....	10
3. Data, Variables and Methodology.....	12
3.1 Data Source and Sample.....	12
3.2 Dependent Variable.....	13
3.3 Main Explanatory Variables	14
3.4 Control Variables	14
3.5 Missing Values and Analytic Sample.....	15
3.6 Model Specification and Survey Estimation.....	16
3.7 Robustness and Diagnostic Checks	17
4. Empirical Results	18
4.1 Descriptive Statistics	18
4.2 Regression Results.....	19
4.3 Native-Born Robustness Model	22
4.4 Diagnostic Summary and Hypothesis Assessment	22
5. Discussion.....	23
5.1 Main Findings in Relation to Previous Research	23
5.2 Interpreting the Controls and Model Fit	24
5.3 Implications	25
5.4 Limitations.....	26
5.5 Directions for Further Research	27
6. Conclusion	27
References	29
Appendix A. Diagnostic Plots.....	32
Appendix B. Declaration of AI Use.....	33

Abstract

Why do some people evaluate immigration more favourably than others? This question matters in Spain because immigration is an important economic, cultural and political issue, while public opinion is not uniform. This thesis compares four established explanations (education, political ideology, economic insecurity and intergroup contact) within one model of Spanish public opinion. It uses the Spanish sample of European Social Survey Round 7 (N = 1,468) and survey-weighted ordinary least squares regression on an index of immigration's perceived economic, cultural and overall effects. More education and more frequent intergroup contact are associated with more positive evaluations, while moving further to the right on the political scale is associated with less positive evaluations. Felt income difficulty follows the expected negative pattern, but the categories are only marginally significant as a group in the full sample and jointly significant among respondents born in Spain. Overall, ideological and social factors are at least as clearly related to the index as economic insecurity. Because the data are cross-sectional, the results show associations rather than causal effects. This caution is especially important for contact because people with more positive attitudes may also be more likely to experience intergroup contact.

Keywords: immigration attitudes; education; political ideology; economic insecurity; intergroup contact; Spain.

1. Introduction

Spain offers a useful setting for studying public attitudes towards immigration because it changed from being mainly a country of emigration to an important country of immigration within a relatively short period (Escandell and Ceobanu, 2009; Cea D’Ancona, 2016). Public views are also politically important because they form part of the setting in which immigration and integration policies are discussed (Ceobanu and Escandell, 2010; Hainmueller and Hopkins, 2014). Spanish evidence shows that material and symbolic evaluations can both matter, and that recent attitude profiles do not follow one identical pattern across impacts, policies and immigrant groups (Moldes-Anaya, Jiménez Aguilar and Jiménez Bautista, 2018; González-Enríquez, Mariscal-de-Gante and Rinken, 2024).

Against this background, the research question is:

To what extent are education, political ideology, economic insecurity, and intergroup contact associated with an immigration-impact attitude index based on perceived economic, cultural and general country impacts of immigration in Spain?

To answer the question, the study uses the Spanish sample from ESS Round 7 and an index of perceived economic, cultural and general country impacts; this round is suitable because its special immigration module contains the impact and contact questions needed for the analysis (European Social Survey European Research Infrastructure, 2023).

This thesis makes two main contributions. First, education, ideology, economic insecurity and contact are studied together in one Spain-specific model. The construction and reliability of the three-item dependent variable are also reported clearly. Second, the study keeps an important mixed result for economic insecurity. The full-sample result is only marginal, while the categories are significant as a group in the native-born model. These two results are reported separately so that the conclusion is not stronger than the evidence.

The rest of the thesis is organised as follows. Section 2 reviews the previous literature and develops the four hypotheses. Section 3 explains the data, variables, missing values and method. Section 4 presents the descriptive statistics and regression results. Section 5 discusses the findings, implications and limitations. Finally, Section 6 presents the conclusion.

2. Literature Review and Hypothesis Development

2.1 Understanding Public Attitudes Towards Immigration

Studies of public responses to immigration do not all use the same outcome. Some studies ask whether a country should admit more immigrants. Other studies examine welfare rights, citizenship, perceived threat, social distance or the economic and cultural effects of immigration. Card, Dustmann and Preston (2005) show that the first ESS immigration module covered beliefs about jobs, wages, welfare, culture, social tension and social distance. Ceobanu and Escandell (2010) also show that comparative research uses several theories and dependent variables. Therefore, a result about admission policy or welfare rights cannot automatically be treated as a result about the perceived effects of immigration.

The outcome used here is more specific than the general term ‘immigration attitudes’. It measures evaluations of the economic, cultural and overall country effects of immigration. These three parts are connected because each asks about a perceived consequence for the receiving country and is coded in the same direction. However, the index does not directly measure immigrant admission, welfare rights, citizenship, prejudice towards one group or every other immigration policy. This boundary matters because attitudes towards immigration and attitudes towards immigrants have several dimensions (Card, Dustmann and Preston, 2005; Ceobanu and Escandell, 2010; Rinken and Mariscal-de-Gante, 2024).

Spain-specific studies also support this careful definition. González-Enríquez, Mariscal de-Gante and Rinken (2024) find five later attitude profiles whose views about impacts, policies and immigrant groups do not always follow the same pattern. The timing of the evidence is important. The Spanish Round 7 interviews were conducted in 2015, while Rinken (2025) studies a later period and finds stronger ideological polarisation, driven substantially by more favourable attitudes on the left. This later development is useful for comparison, but it cannot explain an ideological relationship already present in Round 7.

Four theoretical frameworks organise the analysis. The education or cognitive liberalisation framework connects schooling with knowledge, critical reflection, tolerance and broader social experience (Ceobanu and Escandell, 2010; Hainmueller and Hiscox, 2007). The ideological-identity framework links immigration attitudes with left–right differences concerning national identity, redistribution and group boundaries (McLaren, 2003; Sides and Citrin, 2007). Economic-threat theory focuses on perceived competition for scarce jobs, welfare and other resources (Billiet, Meuleman and De Witte, 2014). Finally, the contact hypothesis expects

interaction across groups to reduce distance and prejudice under favourable conditions (Allport, 1954; Pettigrew and Tropp, 2006).

2.2 Education and Immigration Attitudes

Education is one of the most regular predictors of immigration attitudes in European survey research. Card, Dustmann and Preston (2005) find a clear difference between respondents with lower and higher education in the first ESS immigration module. Rustenbach (2010), Sides and Citrin (2007) and Schneider (2008) also find more favourable attitudes among more educated respondents, although their exact outcomes and models differ. Meuleman, Davidov and Billiet (2009) find a similar education pattern across almost all of the country-round samples in their study of preferences about admitting immigrants.

There are different explanations for this relationship. One economic explanation treats education as a measure of labour-market skills, so lower-skilled native workers may expect more competition from lower-skilled immigrants. However, Hainmueller and Hiscox (2007) find that highly educated Europeans are more supportive of both highly skilled and less-skilled immigrants, and that the education pattern is similar inside and outside the labour force. The authors give more importance to tolerance, cultural openness and beliefs about the national effects of immigration. This is consistent with the cognitive-liberalisation argument rather than a simple job-competition explanation.

Hainmueller and Hopkins (2014) reach a similar conclusion in their review. Personal economic interest is usually a weak explanation, while views about the country and culture are more consistent predictors. Education may influence the values and information people use when they think about immigration, but it is also related to occupation, income, residence and social networks. The education coefficient can therefore reflect more than one process.

Cavaillé and Marshall (2019) provide stronger evidence about the possible effect of education. They use compulsory-schooling reforms in five western European countries and find that an extra year of secondary education reduces anti-immigration attitudes for the people whose schooling was changed by the reforms. Their measures include a question about whether immigration makes the country a worse or better place to live. However, Spain was not included in their causal estimates because the Spanish reform did not have a strong enough statistical relationship with years of education. The study supports the wider education argument, but it is not direct causal evidence for Spain.

Economic and cultural explanations can also work together. Dustmann and Preston (2007) show that labour-market, welfare and racial or cultural concerns have different importance across immigrant groups. Schneider (2008) finds more threat among economically vulnerable respondents, but the country-level results do not support a simple account based only on competition from lower-educated immigrants. Overall, education is expected to remain related to attitudes after economic insecurity and the other factors are included.

Based on this evidence, the first hypothesis expects a positive relationship between years of education and the immigration-impact attitude index, while the other predictors and controls are held constant (H1).

2.3 Political Ideology and Immigration Attitudes

Political ideology can influence how people understand social change, national identity, redistribution and the rights of different groups. Immigration is connected to each of these issues, so left–right self-placement is often related to immigration attitudes (McLaren, 2003; Ceobanu and Escandell, 2010). However, the strength of this relationship can change between countries and over time (Rinken, 2025).

Most cross-national studies used here find less favourable attitudes among respondents on the political right. McLaren (2003) finds that right-wing self-placement is associated with more perceived threat and stronger support for excluding immigrants. Sides and Citrin (2007) find that national identity and cultural unity are more strongly related to European opinion than personal economic dissatisfaction. Rustenbach (2010) also identifies political orientation, education and trust as important predictors. Ceobanu and Escandell (2010) describe political orientation as a repeated individual-level finding, while warning that studies do not always use the same measures.

Research from Spain gives relevant but not identical results. In the Spanish Round 7 analysis by Moldes-Anaya, Jiménez Aguilar and Jiménez Bautista (2018), political orientation was considered but did not remain in their final model, which uses a different outcome and specification. Later Spanish research finds ideological differences between attitude profiles (González-Enríquez, Mariscal-de-Gante and Rinken, 2024). Rinken (2025) also shows that the later widening of the left–right relationship was not a simple one-sided change.

The left–right scale in this thesis is a broad measure from 0 to 10. It does not separately measure party choice, nationalism, authoritarian values or views about one immigration policy.

Its coefficient therefore shows a general ideological relationship rather than one specific reason for it. Immigration attitudes may also influence political identification, so the relationship should not be read as a one-way mechanism (Rinken, 2025).

Based on the previous evidence, the second hypothesis expects that moving further to the right on the political scale will be associated with a less positive immigration-impact score (H2).

2.4 Economic Insecurity and Immigration Attitudes

Economic-threat explanations argue that people may see immigration as increasing competition for jobs, wages, housing, public services or welfare. People who already have financial difficulties may be more concerned about these possible costs (Ceobanu and Escandell, 2010; Dustmann and Preston, 2007). The dependent variable in this thesis also includes a question about the effect of immigration on the economy. Therefore, an association between household income difficulty and the index is reasonable to expect.

However, the evidence is not simple. Billiet, Meuleman and De Witte (2014) study native majority respondents in ESS Round 5 and find some relationships between economic insecurity and ethnic threat, but the individual-level effects are generally modest and the contextual findings are mixed. Heizmann, Jedinger and Perry (2018) find that subjective material deprivation is more consistently related to welfare chauvinism than several objective employment-risk measures. This is relevant because the variable used here asks how people feel about living on their present household income. Nevertheless, welfare chauvinism is not the same outcome as perceived immigration impacts.

Other studies also question a strong explanation based on personal economic interest. Hainmueller and Hopkins (2014) conclude that personal economic circumstances are usually less consistent than views about national economic and cultural effects. Sides and Citrin (2007) find that economic dissatisfaction matters, but that identities and information have stronger relationships. Hainmueller and Hiscox (2007) show that the education pattern does not follow a simple labour-market competition prediction. In the United Kingdom, Dustmann and Preston (2007) also find that welfare and racial or cultural concerns can be more important than direct labour-market concerns.

The evidence from Spain is also mixed. Cea D’Ancona (2016) finds that the economic crisis made material threat more important, but also points to cultural threat, perceived

immigrant presence, political discussion and media attention. Moldes-Anaya, Jiménez Aguilar and Jiménez Bautista (2018) find both economic-material and symbolic factors in the Spanish Round 7 data. Later work also shows that material concerns describe some negative attitude profiles but not all of them (González-Enríquez, Mariscal-de-Gante and Rinken, 2024). This means that economic insecurity may matter without being a complete explanation.

The measurement used in each study is important. Unemployment, occupational risk, national economic conditions, welfare concerns and difficulty living on household income are related ideas, but they are not identical. The outcome also matters because welfare rights, ethnic threat and perceived country impacts measure different attitudes (Ceobanu and Escandell, 2010; Rinken and Mariscal-de-Gante, 2024). Therefore, the economic-insecurity hypothesis is supported by theory, but previous findings make it less certain than the education and ideology hypotheses.

Based on this evidence, the third hypothesis expects greater difficulty living on household income to be associated with a less positive immigration-impact score (H3).

2.5 Intergroup Contact and Immigration Attitudes

Intergroup contact theory argues that interaction between groups can reduce prejudice by increasing knowledge, reducing anxiety and making members of another group less distant. Allport (1954) explains that contact is especially helpful when groups have equal status, common goals, cooperation and support from authorities. Pettigrew and Tropp (2006) review more than 500 studies and find an overall negative relationship between contact and prejudice across many groups and settings. They also find that favourable conditions improve the results, although every condition does not always need to be present.

European survey studies generally find a positive relationship between contact and immigration attitudes, but the type of contact matters. McLaren (2003) finds that friendship with minority-group members is related to lower threat and exclusion. Schneider (2008) also finds more favourable attitudes among people with immigrant friends and, less consistently, other forms of contact. Using ESS Round 7, Neumann and Moy (2018) find stronger results for closer forms such as friendship, while results for contact frequency are more mixed. This suggests that frequency is useful but does not measure the whole contact experience.

Studies from Spain are particularly useful for this thesis. Escandell and Ceobanu (2009) find that close and occasional contact with African and Latin American immigrants is related

to less threat and exclusionism, although workplace contact is not consistent in every model. Wilson-Daily, Kemmelmeier and Prats (2018) find that more frequent cross-group contact among students in Catalonia is related to less xenophobia and more support for diversity and immigrant rights. Their respondents are adolescents and not a national adult sample, so the study gives supporting Spanish evidence but is not directly comparable to this thesis.

The meaning of contact depends on what is measured. Friendship and other close forms of contact are not the same as frequency while out and about, and contact with a person from a different racial or ethnic group does not necessarily mean contact with an immigrant (Neumann and Moy, 2018). Contact can also be positive, negative or neutral. Laurence and Bentley (2018) find opposite relationships for positive and negative interethnic contact. A frequency measure therefore does not capture the quality of the experience.

Another problem is self-selection. People who already have favourable attitudes may choose more diverse friends, workplaces or neighbourhoods. This can make contact and attitudes related even when prior attitudes explain part of the difference. Pettigrew and Tropp (2006), McLaren (2003) and Escandell and Ceobanu (2009) discuss this reverse relationship. The expected coefficient is therefore consistent with contact theory, but it is not by itself evidence that contact produced the attitude difference.

Even with these limitations, the previous evidence gives a clear expectation. Respondents who report more frequent contact with people from a different racial or ethnic group are expected to have a more positive immigration-impact score (H4).

2.6 Summary of Hypotheses

The four hypotheses compare related but different explanations. Education concerns cognitive liberalisation and wider social experience; ideology concerns political identity and group boundaries; economic insecurity concerns perceived material competition; and contact concerns interaction across groups (Hainmueller and Hiscox, 2007; McLaren, 2003; Billiet, Meuleman and De Witte, 2014; Pettigrew and Tropp, 2006). The empirical model estimates these relationships together while the other predictors and controls are held constant.

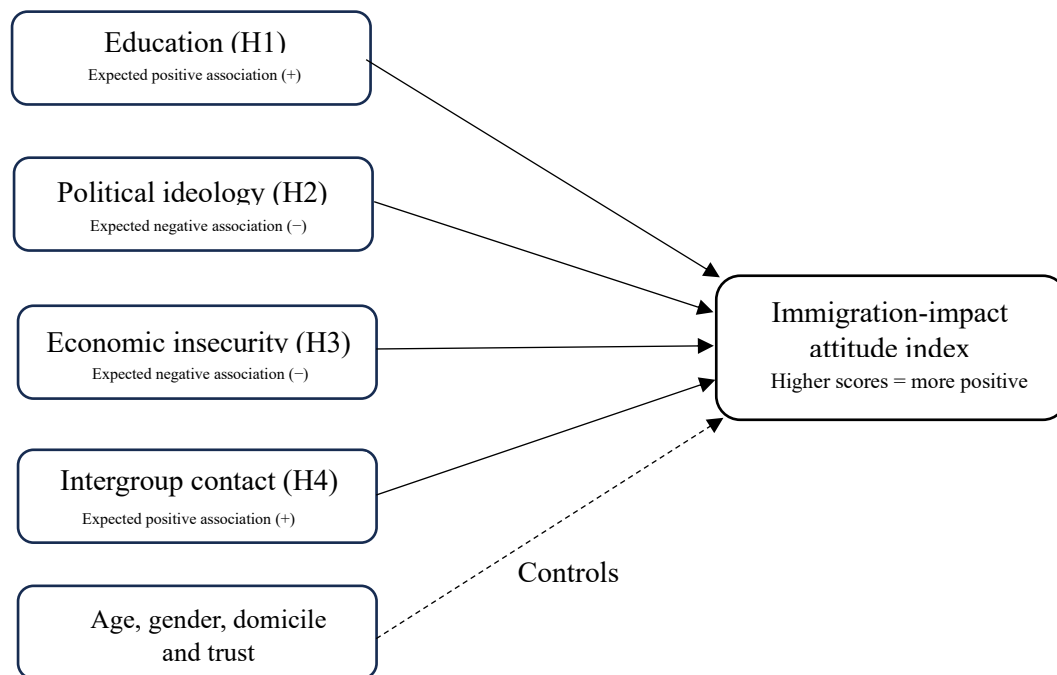
This combined model is useful because earlier studies often focus on one main explanation, another country or a different outcome. The expected directions are clearest for education, ideology and contact, while the previous evidence for personal economic insecurity is more mixed.

Table 1. Summary of Hypotheses

Hypothesis	Expected relationship
H1	Individuals with more years of education are expected to report more positive attitudes towards immigration.
H2	Individuals who place themselves further to the right on the left–right political scale are expected to report less positive attitudes toward immigration.
H3	Individuals who experience greater difficulty living on household income are expected to report less positive attitudes toward immigration.
H4	Individuals who have more frequent contact with people from a different race or ethnic group are expected to report more positive attitudes toward immigration.

Note: The hypotheses concern conditional associations, not causal effects

Figure 1. Conceptual Model of Expected Conditional Associations



Note: Solid arrows show the expected directions of H1–H4. The dashed line identifies the control variables.

3. Data, Variables and Methodology

3.1 Data Source and Sample

The study uses the ESS Round 7 integrated file, edition 2.3 (European Social Survey European Research Infrastructure, 2023). ESS Round 7 is identified as the 2014 round and includes a special module on immigration. The survey covers people aged 15 and over who live in private households. According to the interview-year and interview-month variables in the data, all Spanish interviews were completed between January and June 2015. The dataset is available under DOI 10.21338/ess7e02_3. Spanish fieldwork began on 23 January 2015, shortly after the Charlie Hebdo attack in Paris on 7 January, when the event received intense media coverage across Europe, but this timing is included only as context and is not treated as a causal explanation (Sacko, 2026).

The data are restricted to respondents with the country code ES. This gives an initial unweighted sample of 1,925 respondents. The individual respondent is the unit of analysis. The design is cross-sectional because every respondent is observed once in the same survey round. Therefore, the analysis can identify relationships between variables, but it cannot establish which variable changed first or prove a causal effect.

The main analysis includes respondents born in Spain and respondents born in another country. This represents the resident Spanish sample covered by the ESS. However, country of birth may be related to both immigration attitudes and contact. A robustness model is therefore estimated only for respondents born in Spain. In the original sample, 1,756 respondents were born in Spain and 169 were born elsewhere.

3.2 Dependent Variable

The dependent variable is an immigration-impact attitude index made from three ESS questions (European Social Survey European Research Infrastructure, 2023). The variable *imbgeco* asks whether immigration is bad or good for the country's economy. The variable *imueclt* asks whether immigrants undermine or enrich the country's cultural life. The variable *imwbcnt* asks whether immigrants make the country a worse or better place to live. Each question ranges from 0 to 10, and a higher value is more positive.

The index is calculated as the arithmetic mean of the three questions:

$$att_index_i = \frac{imbgeco_i + imueclt_i + imwbcnt_i}{3}$$

A respondent needs a valid answer to all three questions. This gives every included respondent an index with the same three parts. Taking the mean also keeps the original 0–10 range. A score near 0 means a more negative assessment of the perceived effects of immigration, while a score near 10 means a more positive assessment.

Rustenbach (2010, p. 62) uses the same three impact questions and reports an alpha of 0.81. Schneider (2008) includes the three questions in a broader six-item perceived-threat index and reports a Cronbach's alpha of 0.84 for the pooled sample (p. 57) and 0.81 for Spain (p. 67). Billiet, Meuleman and De Witte (2014) also use the three questions but reverse their coding, so higher values indicate greater threat. These studies support treating the items as a scale.

The three questions work well together in this sample. Cronbach's alpha is 0.8303. The correlations are 0.6138 between *imbgeco* and *imueclt*, 0.6247 between *imbgeco* and *imwbcnt*, and 0.6280 between *imueclt* and *imwbcnt*. The alpha values after removing one question are 0.7673, 0.7725 and 0.7571 respectively. All of them are lower than the full alpha, so no question makes the scale less reliable. In total, 1,715 of the 1,925 Spanish respondents have a valid score on all three questions.

The index is therefore interpreted as a 0–10 measure of perceived economic, cultural and overall country impacts of immigration.

3.3 Main Explanatory Variables

Four explanatory variables are used for the hypotheses. Education is measured by *eduyrs*, which records the completed years of full-time education. It is entered as a continuous variable. A positive coefficient would support H1.

Political ideology is measured by *lrscale*. Respondents place themselves from 0 on the left to 10 on the right. The variable is entered as continuous. A negative coefficient would support H2. This question has more missing answers than the other main predictors, which is explained in Section 3.5.

Economic insecurity is measured by *hincfel*, which asks how the household feels about its present income. The answers are living comfortably, coping, finding it difficult and finding it very difficult. These answers are entered as separate categories because the distance between them may not be equal. Living comfortably is the reference category. H3 expects the coefficients to become more negative when income difficulty increases. The final decision about H3 also uses a joint Wald test of all three included categories.

Intergroup contact is measured by *dfegcon*. This question records how often a respondent has any verbal or non-verbal contact, while out and about, with people of a different race or ethnic group from most people in Spain. It ranges from 1 for never to 7 for every day and is entered as continuous. A positive coefficient would support H4. As explained earlier, the variable measures frequency only. It does not show whether the other person is an immigrant or whether the contact is positive.

3.4 Control Variables

The models include four control variables. Age (*agea*) is measured in years. Gender (*gndr*) is entered as a categorical variable, with male as the reference category. The ESS data for this round contain the categories male and female. Type of residence (*domicil*) separates a big city, the suburbs or outskirts of a big city, a town or small city, a country village, and a farm or home in the countryside. Big city is the reference category. Generalised trust (*ppltrst*) ranges from 0, meaning that one cannot be too careful, to 10, meaning that most people can be trusted (European Social Survey European Research Infrastructure, 2023).

These controls were chosen before estimating the final model because earlier studies include or connect age, gender, residence and generalised trust with immigration attitudes (Sides and Citrin, 2007; Schneider, 2008; Meuleman, Davidov and Billiet, 2009; Rustenbach, 2010). They are not the main hypotheses. For this reason, their coefficients are reported, but unexpected control results are not given a new explanation without evidence.

Table 2. Variables Included in the Empirical Analysis

Variable	Role	Measurement	Treatment in the model
<i>att_index</i>	Dependent variable	Mean of <i>imbgeco</i> , <i>imueclt</i> and <i>imwbcnt</i> , 0–10	Continuous
<i>edyrs</i>	Main explanatory variable, H1	Years of full-time education completed	Continuous
<i>lrscale</i>	Main explanatory variable, H2	Left-right self-placement, 0–10	Continuous
<i>hincfel</i>	Main explanatory variable, H3	Feeling about household income nowadays	Categorical
<i>dfegcon</i>	Main explanatory variable, H4	Frequency of contact with people of a different race or ethnic group, 1–7	Continuous
<i>agea</i>	Control variable	Age in years	Continuous
<i>gndr</i>	Control variable	Gender	Categorical
<i>domicil</i>	Control variable	Domicile/residence type	Categorical
<i>ppltrst</i>	Control variable	Generalized trust, 0–10	Continuous
<i>pspwght</i>	Survey weight	Post-stratification weight including design weight	Weight only

Note: Variable names and coding follow ESS Round 7. Special missing-value codes are recoded before analysis.

3.5 Missing Values and Analytic Sample

ESS uses special numbers for answers such as refusal, don't know and no answer. These numbers are changed to missing before the analysis. The codes are not the same for every

variable. For several 0–10 or numeric variables, the missing codes are 77, 88 and 99. For *hincfel* and *domicil*, the codes are 7, 8 and 9; gender uses 9; and age uses 999. The recoding was checked one variable at a time. Importantly, the value 7 was kept for *dfegcon* because it is a valid answer meaning ‘every day’ (European Social Survey European Research Infrastructure, 2023).

The regressions use complete cases. Model 1 and Model 2 are estimated with the same 1,468 respondents. This allows the coefficients and R^2 values to be compared without also changing the sample. The decrease from 1,925 respondents is caused by missing values in the index and the predictors. The largest single source is political self-placement: 263 respondents, or 13.7% of the original sample, do not have a valid *lrscle* answer. Missing values are not imputed.

Complete-case analysis is easy to reproduce and does not create values that were not observed. However, it can create bias if respondents with complete answers are systematically different from respondents with missing answers. Therefore, the final analytic sample cannot be assumed to be exactly the same as the original Spanish sample. All descriptive statistics and main regressions use the same retained cases and the ESS weight. Table 3 shows how the different samples were formed.

Table 3. Construction of the Analytic Samples

Sample stage	Unweighted N
Initial Spain sample	1,925
Valid three-item immigration-impact index	1,715
Complete-case main analytic sample	1,468
Native-born robustness sample	1,346

Note: No values are imputed. Models 1 and 2 use the same 1,468 respondents.

3.6 Model Specification and Survey Estimation

Ordinary least squares (OLS) regression is used. The dependent variable is the mean of three 0–10 questions and has 31 possible values, so it can be treated as approximately continuous for this analysis. OLS also allows the coefficients to be read directly as points on the 0–10 index. The full model is written as follows:

$$Attitude_i = \beta_0 + \beta_1 Education_i + \beta_2 Ideology_i + \sum_k \beta_{3k} Income_{ik} + \beta_4 Contact_i + \gamma' Controls_i + \varepsilon_i$$

Model 1 includes only age, gender, domicile and generalized trust. Model 2 adds education, political ideology, the three income-difficulty indicators and contact frequency. Both models use the same observations. This makes it possible to see how much more variation is explained after adding the four main predictors. Model 3 repeats the full model only for respondents born in Spain.

All reported descriptive statistics and regression estimates use the ESS post-stratification weight *pspwght*. The ESS codebook states that this weight already includes the design weight (European Social Survey European Research Infrastructure, 2023). Since this is a one-country analysis, the population-size weight is not used. In Stata, the survey design is declared with `svyset [pweight=pspwght]`, and the models are estimated with the `svy:` prefix. The table therefore reports linearised survey standard errors and unweighted sample sizes.

H1, H2 and H4 are assessed from the direction, size and statistical uncertainty of their coefficients. H3 is assessed from the pattern of all three income categories and their joint Wald test. A further joint Wald test is used for the domicile categories. Significance is marked at $p < 0.05$, $p < 0.01$ and $p < 0.001$. A p-value between 0.050 and 0.100 is described as marginal evidence and not conventional statistical significance.

3.7 Robustness and Diagnostic Checks

The native-born model checks whether the main pattern remains when respondents born outside Spain are removed. It is a robustness check rather than a formal test that the coefficients differ between native-born and foreign-born respondents.

Multicollinearity is assessed with variance inflation factors because some predictors may be related to one another. Stata does not calculate VIFs after survey estimation, so the same full specification is estimated without survey adjustment for this auxiliary check. A residual histogram and a residual-versus-fitted plot from the auxiliary regression are also examined because the dependent variable is bounded and discrete. The reported coefficients and standard errors continue to come from the survey-weighted models.

4. Empirical Results

4.1 Descriptive Statistics

Table 4 gives the descriptive statistics for the 1,468 respondents used in the main regressions. The means, standard deviations and percentages are survey-weighted. The sample size shown in the table is unweighted.

Table 4. Descriptive Statistics for the Main Analytic Sample

Variable	Mean / Share	Standard deviation
Continuous variables		
Immigration-impact attitude index (<i>att_index</i>)	5.58	1.98
Years of education (<i>edyrs</i>)	13.86	5.76
Left-right political scale (<i>lrscale</i>)	4.38	2.08
Intergroup contact (<i>dfegcon</i>)	4.75	2.18
Generalized trust (<i>ppltrst</i>)	5.00	2.02
Age (<i>agea</i>)	46.42	17.37
Categorical variables		
Male	51.01%	
Female	48.99%	
Living comfortably on present income	33.38%	
Coping on present income	44.39%	
Difficult on present income	16.79%	
Very difficult on present income	5.44%	
Big city	19.10%	
Suburbs or outskirts of big city	7.65%	
Town or small city	32.56%	
Country village	37.93%	
Farm or home in countryside	2.76%	

Note: Weighted means, weighted percentages and weighted standard deviations. Unweighted N = 1,468.

The mean immigration-impact score is 5.58 on the 0–10 scale. This is slightly above the midpoint. The standard deviation is 1.98, which shows that respondents give quite different answers. The mean number of years of full-time education is 13.86. The average left–right position is 4.38, which is a little to the left of the midpoint. Mean contact frequency is 4.75 on the 1–7 scale, generalised trust is 5.00, and mean age is 46.42 years.

The weighted sample is 51.01% male and 48.99% female. For household income, 33.38% say that they live comfortably and 44.39% say that they are coping. Another 16.79% find it difficult and 5.44% find it very difficult. Most respondents in the analytic sample live in a town or small city (32.56%) or a country village (37.93%). Big-city residents make up 19.10%, suburban residents 7.65%, and respondents living on a farm or in a countryside home 2.76%.

These numbers describe the weighted complete-case sample. They should not be read as unconditional totals for the whole Spanish population because respondents with missing political self-placement or another model variable are not included.

4.2 Regression Results

Table 5 presents the survey-weighted OLS results. Model 1 contains the control variables. Model 2 is the main full model, and Model 3 repeats the full model for respondents born in Spain. The coefficients show changes in points on the immigration-impact index. Linearised standard errors are given in parentheses.

Table 5. Survey-Weighted OLS Models of Immigration-Impact Attitudes

	(1) Controls only	(2) Full model	(3) Native-born only
Years of education		0.056*** (0.009)	0.062*** (0.010)
Left-right self-placement (0-10)		-0.109*** (0.026)	-0.127*** (0.027)
Coping on present income		-0.247* (0.110)	-0.288* (0.114)
Difficult on present income		-0.284 (0.151)	-0.382* (0.158)
Very difficult on present income		-0.454 (0.272)	-0.674* (0.310)
Intergroup contact frequency (1-7)		0.099*** (0.023)	0.090*** (0.024)
Age	-0.014*** (0.003)	-0.002 (0.003)	0.002 (0.003)
Female	-0.549*** (0.100)	-0.537*** (0.097)	-0.578*** (0.101)
Suburbs or outskirts of a big city	0.018 (0.224)	0.064 (0.215)	0.055 (0.222)
Town or small city	-0.219 (0.149)	-0.169 (0.143)	-0.147 (0.149)
Country village	-0.653*** (0.148)	-0.491*** (0.145)	-0.497*** (0.150)
Farm or home in the countryside	-0.992*** (0.291)	-0.706* (0.280)	-0.805** (0.297)
Generalized trust (0-10)	0.175*** (0.026)	0.133*** (0.026)	0.142*** (0.026)
Constant	5.956*** (0.250)	4.938*** (0.348)	4.694*** (0.367)
Observations	1,468	1,468	1,346
R ²	0.097	0.153	0.170

*Note: Survey-weighted OLS; linearised standard errors in parentheses. Reference categories: male, big city and living comfortably on present income. Joint Wald tests — income: Model 2, $F(3,1465) = 2.34$, $p = 0.072$; Model 3, $F(3,1343) = 3.55$, $p = 0.014$. Domicile in Model 2: $F(4,1464) = 4.92$, $p = 0.001$. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.*

The controls-only model explains 9.75% of the variation in the index. After the four main predictors are added, R^2 increases from 0.0975 to 0.1530, an increase of 0.0555 or about 5.6 percentage points. Models 1 and 2 use the same 1,468 respondents, so this comparison is not caused by a change in sample. Most of the variation nevertheless remains unexplained.

Education has the expected positive coefficient. In Model 2, one more year of education is associated with a 0.056-point higher immigration-impact score ($SE = 0.009$, $p < 0.001$), with the other variables held constant. A difference of four years is equal to about 0.224 index points.

Political ideology has the expected negative coefficient. Moving one point to the right is associated with a 0.109-point lower score ($SE = 0.026$, $p < 0.001$). A five-point difference on the political scale is equal to about 0.545 index points, with the other variables held constant.

All three household-income coefficients are negative compared with living comfortably. Respondents who are coping score 0.247 points lower ($SE = 0.110$, $p = 0.025$). The coefficient is -0.284 for respondents finding it difficult ($SE = 0.151$, $p = 0.060$) and -0.454 for those finding it very difficult ($SE = 0.272$, $p = 0.095$). The order follows H3, but the joint test gives $F(3,1465) = 2.34$ and $p = 0.072$. The estimates for the two most difficult categories are also less precise, and Table 4 shows that these groups represent only 16.8% and 5.4% of the weighted sample.

Contact frequency has a positive coefficient. A one-point increase on the 1–7 scale is associated with a 0.099-point higher immigration-impact score ($SE = 0.023$, $p < 0.001$). The difference between the two ends of the scale is therefore about 0.594 points, with the other variables held constant.

Using the standard deviations in Table 4, a one-standard-deviation difference corresponds to about 0.32 index points for education, 0.23 for ideology, 0.22 for contact and 0.27 for trust. The four continuous predictors therefore have broadly similar conditional magnitudes, although their directions differ.

Among the controls, age is negative and significant in Model 1 but is close to zero in Model 2. The female coefficient remains negative in both models. Village and farm or countryside

residence have negative coefficients relative to a big city, while generalised trust remains positive. The suburb and town coefficients are not statistically significant.

4.3 Native-Born Robustness Model

Model 3 includes 1,346 respondents who were born in Spain. Education, ideology and contact keep the same direction and remain statistically significant. The education coefficient is 0.062 (SE = 0.010, $p < 0.001$), the ideology coefficient is -0.127 (SE = 0.027, $p < 0.001$), and the contact coefficient is 0.090 (SE = 0.024, $p < 0.001$). These estimates are close to the full-sample results.

Household-income difficulty is more clearly related to the index in Model 3. Compared with living comfortably, the coefficients are -0.288 for coping ($p = 0.011$), -0.382 for finding it difficult ($p = 0.016$), and -0.674 for finding it very difficult ($p = 0.030$). The joint test gives $F(3,1343) = 3.55$ and $p = 0.014$. This model does not formally test whether the coefficients differ between native-born and foreign-born respondents. Model 3 has an R^2 of 0.170.

4.4 Diagnostic Summary and Hypothesis Assessment

The VIF results do not show a serious multicollinearity problem. The mean VIF is 1.32 and the highest value is 2.01. The residual plots show the banded pattern expected from an index made from three bounded integer questions and some narrowing near the fitted-value limits. The survey-weighted OLS models and linearised standard errors are retained, while the bounded dependent variable remains a limitation. The two diagnostic plots are reproduced in Appendix A. Table 6 gives the final assessment of the four hypotheses.

Table 6. Hypothesis Assessment

Hypothesis	Expected	Verdict	Key evidence
H1	Education: positive	Supported	Model 2: $b = 0.056$, $p < 0.001$; Model 3: $b = 0.062$, $p < 0.001$
H2	Rightward ideology: negative	Supported	Model 2: $b = -0.109$, $p < 0.001$; Model 3: $b = -0.127$, $p < 0.001$
H3	Income difficulty: negative	Marginal in full sample; supported among native-born respondents	Joint $p = 0.072$ in Model 2; joint $p = 0.014$ in Model 3
H4	Contact: positive	Supported as an association	Model 2: $b = 0.099$, $p < 0.001$; Model 3: $b = 0.090$, $p < 0.001$

Note: Verdicts refer to conditional associations in the estimated survey-weighted models.

5. Discussion

5.1 Main Findings in Relation to Previous Research

Table 6 shows that education, political ideology and intergroup contact have clear relationships with the immigration-impact index. Economic insecurity follows the expected negative order, but the strength of the evidence differs between the full and native-born samples. The findings therefore do not support one single explanation of immigration attitudes.

The positive education result agrees with a common finding in European research. Hainmueller and Hiscox (2007), Schneider (2008), Rustenbach (2010) and Meuleman, Davidov and Billiet (2009) all report more favourable attitudes among more educated respondents, although their outcomes and models differ. Cavaillé and Marshall (2019) also find that extra secondary schooling reduces anti-immigration attitudes in the countries included in their reform analysis, while Spain was not included in those causal estimates. The present coefficient may reflect values, knowledge, occupation, networks and selection into schooling,

but it remains positive after income difficulty, ideology, contact, domicile and trust are included.

The ideology result also agrees with previous European and Spanish research (McLaren, 2003; Sides and Citrin, 2007; Rustenbach, 2010; González-Enríquez, Mariscal-de-Gante and Rinken, 2024). The size is meaningful: a five-point difference is associated with about half an index point. However, *lrscale* is a broad political position and not an explanation by itself. Rinken's (2025) later analysis also shows that Spanish polarisation changed over time, so the Round 7 coefficient should not be read as a result of later events.

The income result fits the mixed findings in earlier research. Its negative order is consistent with a link between subjective material difficulty and less favourable attitudes (Heizmann, Jedinger and Perry, 2018). The full-sample evidence is only marginal, while the categories are jointly significant among respondents born in Spain. This pattern is consistent with group-threat accounts in which majority-group members who feel economically vulnerable may perceive competition over scarce resources (Billiet, Meuleman and De Witte, 2014). It is not a formal test that nativity changes the relationship, and it should not be taken as proof of the theory.

The positive contact coefficient is consistent with Allport's (1954) argument and the review by Pettigrew and Tropp (2006). It also agrees with European evidence from McLaren (2003), Schneider (2008) and Neumann and Moy (2018), as well as Spanish evidence from Escandell and Ceobanu (2009) and Wilson-Daily, Kemmelmeier and Prats (2018). However, frequency does not identify an immigrant contact or show whether the experience was positive, negative or neutral (Laurence and Bentley, 2018). Prior attitudes may also influence who experiences more intergroup contact.

5.2 Interpreting the Controls and Model Fit

Generalised trust has a positive and stable coefficient. This agrees with studies that find more favourable immigration attitudes among respondents with higher social trust (Sides and Citrin, 2007; Rustenbach, 2010). Because the trust question does not mention immigration, the result suggests that evaluations of immigration are connected to wider views about other people.

Village residents and people living on farms or in countryside homes have lower scores than big-city residents after the other variables are included. Domicile may be related to social

environments, local politics, information or opportunities for contact. However, it does not measure regional location, local immigrant shares or neighbourhood diversity, so the model cannot show which explanation is correct.

Age is negative in Model 1 but becomes close to zero after education and the main predictors are added. This is consistent with the controls-only age result partly representing cohort differences in schooling, but one cross-section cannot separate age, period and cohort or prove that explanation. The female coefficient also remains negative and is close to -0.54 in the full model. Since this was not a hypothesis and earlier studies do not show one consistent gender pattern, it is reported as a conditional difference rather than given a new explanation after seeing the result (Meuleman, Davidov and Billiet, 2009; Moldes-Anaya, Jiménez Aguilar and Jiménez Bautista, 2018).

The full-model R^2 is 0.153, leaving most of the observed variation unexplained. Previous research points to other possible factors such as national identity, political context, information, images of immigrant groups and the quality of contact (Sides and Citrin, 2007; Ceobanu and Escandell, 2010; Laurence and Bentley, 2018). The purpose of the model is to compare four important relationships clearly, not to predict every difference in immigration attitudes.

5.3 Implications

The results have several careful implications. First, the education coefficient is difficult to explain only through direct competition for jobs. Hainmueller and Hiscox (2007) and Hainmueller and Hopkins (2014) give more importance to tolerance, knowledge and views about national consequences. This suggests that education and access to reliable information are relevant to how immigration is understood. The analysis does not evaluate an education or information programme, so it cannot estimate the effect of a particular intervention.

Second, the ideology coefficient shows that people with different political positions can interpret the same issue differently. Information may therefore not create the same response among everyone when immigration is connected to identity and political conflict (Sides and Citrin, 2007; Rinken, 2025).

Third, economic difficulty should not be ignored, but it should not be treated as the only reason for immigration attitudes. The native-born result is consistent with group-threat reasoning, while the full-sample evidence remains marginal. Policies that reduce material insecurity may be important for many reasons, but these results do not estimate their effect on

immigration attitudes (Billiet, Meuleman and De Witte, 2014; Hainmueller and Hopkins, 2014).

Finally, more frequent contact is associated with more positive attitudes, but more contact is not always better contact. Equal status, cooperation, common goals and institutional support matter in contact theory (Allport, 1954), while positive and negative experiences can be related to attitudes in opposite directions (Laurence and Bentley, 2018). The quality and setting of contact may therefore matter as much as its frequency.

5.4 Limitations

The first and most important limitation is the cross-sectional design. Education, ideology, income feeling, contact and attitudes are measured in one survey round. Unmeasured variables may affect the coefficients, and influence may work in both directions. Causal statements are therefore not possible. Reverse causality is especially important for contact because more tolerant people may select into more diverse relationships and environments (Pettigrew and Tropp, 2006; Escandell and Ceobanu, 2009).

Second, complete-case analysis reduces the sample from 1,925 to 1,468 respondents. Missing answers are especially common for *lrscale*, where 263 respondents, or 13.7% of the original sample, do not have a valid answer. The ESS weight corrects the features for which it was designed, but it cannot guarantee that missing answers on the model variables are corrected. If people who do not report an ideology also have different immigration attitudes, the final sample may not represent them correctly.

Third, the dependent variable is limited to values from 0 to 10 and is the average of three ordinal survey questions. OLS gives coefficients that are easy to understand, but it assumes a linear conditional relationship and can theoretically predict values outside the scale. The index also gives equal weight to the economic, cultural and overall questions. The good Cronbach's alpha shows that the questions work well together. It does not mean that the three parts have exactly the same meaning.

Fourth, the variables do not measure every part of the concepts. *hincfel* measures how respondents feel about living on household income, but not wealth, exact income, unemployment risk or expectations. *dfegcon* measures how often respondents have verbal or non-verbal contact with someone from a different racial or ethnic group while out and about, but not whether that person is an immigrant or whether the experience is positive. *lrscale* places several political ideas on one line, and *domicil* does not measure the number of immigrants

living nearby (European Social Survey European Research Infrastructure, 2023; Rinken and Mariscal-de-Gante, 2024).

Fifth, the native-born analysis is a robustness check and not a complete subgroup analysis. It shows that the main coefficients keep their direction and that the income categories are jointly significant among Spain-born respondents. However, it does not formally show that the relationships are different for foreign-born respondents. The foreign-born group is also small, which would make a separate comparison less precise.

Finally, the study concerns Spain at the time of the 2015 Round 7 interviews. The results should not be applied without qualification to other countries, every Spanish region or later political periods. Immigration flows, party competition and the public importance of the issue can change over time (Cea D’Ancona, 2016; Rinken, 2025). In the same way, findings from later Spain should not be treated as if they already described the Round 7 respondents.

5.5 Directions for Further Research

Future research could use panel or longitudinal data to study which variable changes first, especially for contact and ideology. Repeated cross-sectional surveys could show whether the coefficients change when immigration becomes more politically important. A multilevel study could add regional unemployment, immigrant shares and political context, helping to separate individual economic difficulty from conditions in the place where a respondent lives (Ceobanu and Escandell, 2010; Meuleman, Davidov and Billiet, 2009).

The measures could also be improved. Separate models for the economic, cultural and overall questions could show whether the predictors are related differently to each part. Contact questions could separate positive and negative experiences, friendship, workplace contact and neighbourhood contact (Neumann and Moy, 2018; Laurence and Bentley, 2018). Economic insecurity could combine household-income feelings with employment, income and expectations. Finally, interaction tests could examine whether income difficulty or contact has different relationships by nativity, education, ideology or local context.

6. Conclusion

This thesis asked how education, political ideology, economic insecurity and intergroup contact are associated with perceived immigration impacts in Spain. The analysis used survey-weighted OLS models for the Spanish sample of ESS Round 7. Table 6 summarises the hypothesis assessment: the evidence is clear for education, ideology and contact, while the

result for economic insecurity is more qualified and differs between the full and native-born samples.

The findings suggest that educational, political, material and social-contact factors all have a place in understanding Spanish public opinion, but no single explanation is sufficient. The model also shows why the mixed income result should be kept rather than simplified. Its negative pattern is visible in the full sample, while the statistical evidence is clearer among respondents born in Spain.

These conclusions remain limited by the complete-case sample, the broad survey measures and the bounded outcome. The language of association is also essential, especially for contact, because prior attitudes may partly shape who experiences intergroup contact. Within these limits, the thesis provides a clear Spain-specific comparison of four established explanations and distinguishes where the evidence is strong from where it remains uncertain.

References

- Allport, G.W. (1954) *The Nature of Prejudice*. Cambridge, MA: Addison-Wesley.
- Billiet, J., Meuleman, B. and De Witte, H. (2014) 'The relationship between ethnic threat and economic insecurity in times of economic crisis: Analysis of European Social Survey data', *Migration Studies*, 2(2), pp. 135–161. <https://doi.org/10.1093/migration/mnu023>
- Card, D., Dustmann, C. and Preston, I. (2005) *Understanding Attitudes to Immigration: The Migration and Minority Module of the First European Social Survey*. CReAM Discussion Paper No. 03/05. London: Centre for Research and Analysis of Migration, University College London.
- Cavaillé, C. and Marshall, J. (2019) 'Education and anti-immigration attitudes: Evidence from compulsory schooling reforms across Western Europe', *American Political Science Review*, 113(1), pp. 254–263. <https://doi.org/10.1017/S0003055418000588>
- Cea D'Ancona, M.Á. (2016) 'Immigration as a threat: Explaining the changing pattern of xenophobia in Spain', *Journal of International Migration and Integration*, 17(2), pp. 569–591. <https://doi.org/10.1007/s12134-015-0415-3>
- Ceobanu, A.M. and Escandell, X. (2010) 'Comparative analyses of public attitudes toward immigrants and immigration using multinational survey data: A review of theories and research', *Annual Review of Sociology*, 36, pp. 309–328. <https://doi.org/10.1146/annurev.soc.012809.102651>
- Dustmann, C. and Preston, I.P. (2007) 'Racial and economic factors in attitudes to immigration', *The B.E. Journal of Economic Analysis & Policy*, 7(1), Article 62. <https://doi.org/10.2202/1935-1682.1655>
- Escandell, X. and Ceobanu, A.M. (2009) 'When contact with immigrants matters: Threat, interethnic attitudes and foreigner exclusionism in Spain's Comunidades Autónomas', *Ethnic and Racial Studies*, 32(1), pp. 44–69. <https://doi.org/10.1080/01419870701846924>
- European Social Survey European Research Infrastructure (2023) ESS7 – Integrated File, Edition 2.3 [data set]. Sikt – Norwegian Agency for Shared Services in Education and Research. https://doi.org/10.21338/ess7e02_3

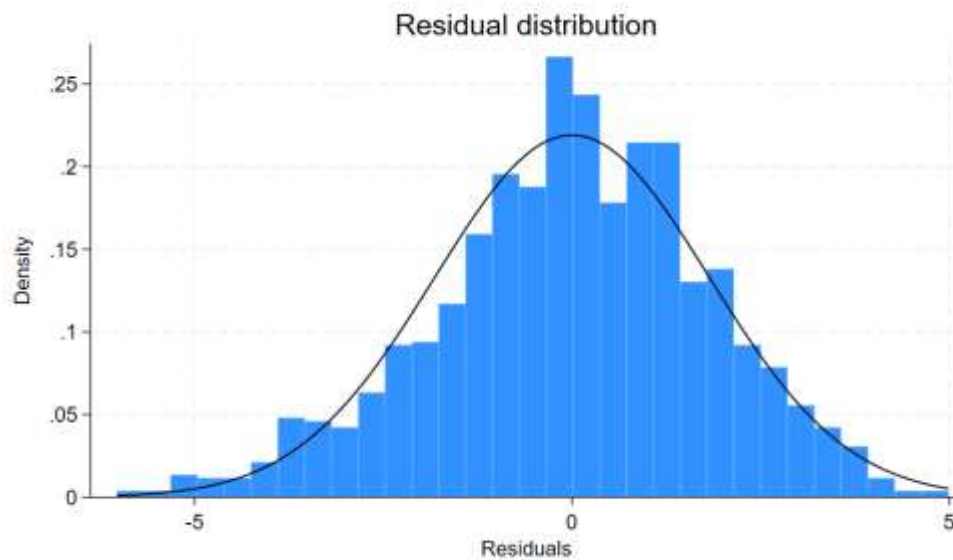
- González-Enríquez, C., Mariscal-de-Gante, Á. and Rinken, S. (2024) 'Attitudes towards immigration and immigrants in Spain: A typology', *Papers. Revista de Sociologia*, 109(3), e3261, pp. 1–25. <https://doi.org/10.5565/rev/papers.3261>
- Hainmueller, J. and Hiscox, M.J. (2007) 'Educated preferences: Explaining attitudes toward immigration in Europe', *International Organization*, 61(2), pp. 399–442. <https://doi.org/10.1017/S0020818307070142>
- Hainmueller, J. and Hopkins, D.J. (2014) 'Public attitudes toward immigration', *Annual Review of Political Science*, 17, pp. 225–249. <https://doi.org/10.1146/annurev-polisci-102512-194818>
- Heizmann, B., Jedinger, A. and Perry, A. (2018) 'Welfare chauvinism, economic insecurity and the asylum seeker “crisis”', *Societies*, 8(3), Article 83. <https://doi.org/10.3390/soc8030083>
- Laurence, J. and Bentley, L. (2018) 'Countervailing contact: Community ethnic diversity, anti-immigrant attitudes and mediating pathways of positive and negative inter-ethnic contact in European societies', *Social Science Research*, 69, pp. 83–110. <https://doi.org/10.1016/j.ssresearch.2017.09.007>
- McLaren, L.M. (2003) 'Anti-immigrant prejudice in Europe: Contact, threat perception, and preferences for the exclusion of migrants', *Social Forces*, 81(3), pp. 909–936. <https://doi.org/10.1353/sof.2003.0038>
- Meuleman, B., Davidov, E. and Billiet, J. (2009) 'Changing attitudes toward immigration in Europe, 2002–2007: A dynamic group conflict theory approach', *Social Science Research*, 38(2), pp. 352–365. <https://doi.org/10.1016/j.ssresearch.2008.09.006>
- Moldes-Anaya, S., Jiménez Aguilar, F. and Jiménez Bautista, F. (2018) 'Actitudes hacia la inmigración en España a través de la Encuesta Social Europea', *OBETS. Revista de Ciencias Sociales*, 13(1), pp. 93–119. <https://doi.org/10.14198/OBETS2018.13.1.04>
- Neumann, R. and Moy, P. (2018) 'You're (not) welcome: The impact of symbolic boundaries, intergroup contact, and experiences with discrimination on immigration attitudes', *American Behavioral Scientist*, 62(4), pp. 458–477. <https://doi.org/10.1177/0002764218760370>

- Pettigrew, T.F. and Tropp, L.R. (2006) 'A meta-analytic test of intergroup contact theory', *Journal of Personality and Social Psychology*, 90(5), pp. 751–783. <https://doi.org/10.1037/0022-3514.90.5.751>
- Rinken, S. (2025) 'La politización de la migración internacional en España: ¿Excepcionalismo duradero?', *Revista CIDOB d'Afers Internacionals*, 140, pp. 79–100. <https://doi.org/10.24241/rcai.2025.140.2.79>
- Rinken, S. and Mariscal-de-Gante, Á. (2024) 'Do dependents matter? Exploring multiple dimensions of attitudes toward immigration and immigrants', *Journal of Ethnic and Migration Studies*, 50(16), pp. 3959–3980. <https://doi.org/10.1080/1369183X.2024.2361072>
- Rustenbach, E. (2010) 'Sources of negative attitudes toward immigrants in Europe: A multi-level analysis', *International Migration Review*, 44(1), pp. 53–77. <https://doi.org/10.1111/j.1747-7379.2009.00798.x>
- Sacko, M. (2026) 'Media narratives and public attitudes toward immigrants and Muslims: Evidence from the Charlie Hebdo attack', *European Journal of Political Economy*, 93, 102812. <https://doi.org/10.1016/j.ejpoleco.2026.102812>
- Schneider, S.L. (2008) 'Anti-immigrant attitudes in Europe: Outgroup size and perceived ethnic threat', *European Sociological Review*, 24(1), pp. 53–67. <https://doi.org/10.1093/esr/jcm034>
- Sides, J. and Citrin, J. (2007) 'European opinion about immigration: The role of identities, interests and information', *British Journal of Political Science*, 37(3), pp. 477–504. <https://doi.org/10.1017/S0007123407000257>
- Wilson-Daily, A.E., Kemmelmeier, M. and Prats, J. (2018) 'Intergroup contact versus conflict in Catalan high schools: A multilevel analysis of adolescent attitudes toward immigration and diversity', *International Journal of Intercultural Relations*, 64, pp. 12–28. <https://doi.org/10.1016/j.ijintrel.2018.03.002>

Appendix A. Diagnostic Plots

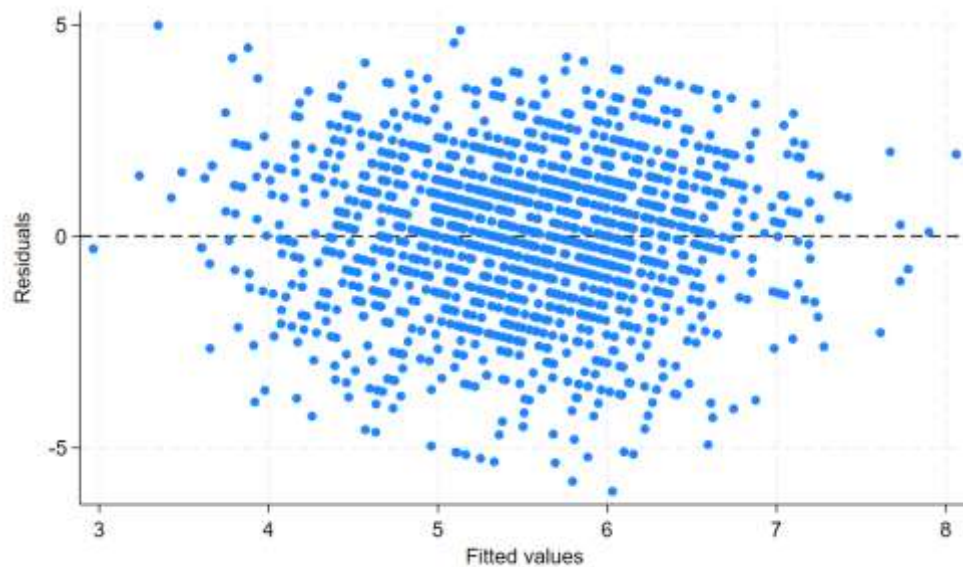
The plots below come from the unweighted auxiliary full specification used only for the diagnostic checks. The coefficients and standard errors reported in the thesis continue to come from the survey-weighted models.

Figure A1. Distribution of Residuals from the Auxiliary Full Specification



Note: Residuals are from the unweighted auxiliary full specification used only for diagnostics.

Figure A2. Residuals versus Fitted Values from the Auxiliary Full Specification



Note: Residuals and fitted values are from the unweighted auxiliary full specification used only for diagnostics.

Appendix B. Declaration of AI Use

AI was used as a technical sounding board to assist with drafting and reviewing Stata .do-file syntax, clarifying regression outputs, and reviewing the implementation and interpretation of diagnostic checks. It was also used to proofread independently drafted text, correct grammatical and idiom errors, and enhance academic tone and structural flow across chapters.

The tools used were Gemini and ChatGPT. All AI-assisted outputs were critically reviewed by the author, who retained responsibility for the analysis, interpretation and final text. The complete protocol, exact prompts and AI responses are submitted separately as supplementary documentation.